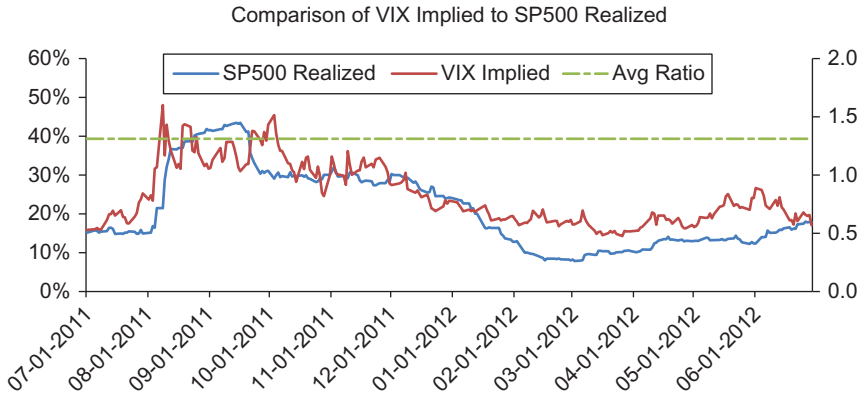




■ **Figure 6.4** Comparison of VIX Implied Volatility to the SP500 Realized Volatility.

place. Just the worry of a potential event taking place will be reflected in the HMA-VIX model. Models updated nightly will miss this event and will not necessarily provide timely accurate volatility estimates.

Determining Parameters via Maximum Likelihood Estimation

Parameters of the GARCH and EWMA volatility models are computed via maximum likelihood estimation (MLE). An overview of the estimation process follows.

Likelihood Function

Let log price returns be normally distributed with mean zero and time varying volatility, that is, $y_t \sim N(0, \hat{\sigma}_t^2)$. Then the probability density function (pdf) of these returns at any time is:

$$f_t(y_t; \hat{\sigma}_t) = \frac{1}{\sqrt{2\pi\hat{\sigma}_t^2}} \cdot e^{-\frac{y_t^2}{2\hat{\sigma}_t^2}} \quad (6.23)$$

The likelihood of achieving the observed series of returns is:

$$L = \prod_{t=1}^n \frac{1}{\sqrt{2\pi\hat{\sigma}_t^2}} \cdot e^{-\frac{y_t^2}{2\hat{\sigma}_t^2}} \quad (6.24)$$